US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12398845
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	West; Nicholas et al.

Valve actuation mechanism, valve arrangement and pipe installation for LH2 applications as well as use thereof

Abstract

A valve actuation mechanism for actuating a rotatable closing valve element for cryogenic applications and/or H2 applications, including a housing defining an inner area sealed by a membrane from an outer area, an input shaft configured to receive a rotational movement, the input shaft being in the outer area, an output shaft configured to be connected to the closing element, wherein the output shaft is in the inner area, and a strain wave gearing connecting the input and output shafts. The strain wave gearing includes a wave generator connected to one of the input and output shafts, a rotatable outer gear wheel connected to the other one of the input and output shafts and a stationary flexible ring gear with external teeth deformed by the wave generator to engage with internal gear teeth of the outer gear wheel. The flexible ring gear is a portion of the sealing membrane.

Inventors: **West; Nicholas (Filton Bristol, GB), Lammert; Nils (Hamburg, DE)**

Applicant: **Airbus Operations GmbH** (Hamburg, DE); **Airbus Operations Limited** (Filton Bristol, GB)

Family ID: **1000008782793**

Assignee: **Airbus Operations GmbH (Hamburg, DE); Airbus Operations Limited (Filton Bristol, GB)**

Appl. No.: **18/481392**

Filed: **October 05, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240142062 A1	May. 02, 2024

Foreign Application Priority Data

EP	22204327	Oct. 28, 2022

Publication Classification

Int. Cl.: **F17C13/04** (20060101); **F16H49/00** (20060101); **F16K1/226** (20060101); **F16K5/06** (20060101); **F16K31/524** (20060101); **F16K31/53** (20060101); **F16K31/60** (20060101); B64D37/30 (20060101)

U.S. Cl.:

CPC **F17C13/04** (20130101); **F16H49/001** (20130101); **F16K1/2268** (20130101); **F16K5/0694** (20130101); **F16K31/524** (20130101); **F16K31/53** (20130101); **F16K31/535** (20130101); **F16K31/60** (20130101); B64D37/30 (20130101)

Field of Classification Search

CPC: F17C (13/04); F16H (49/001); F16H (2049/003); F16K (31/53); F16K (31/524); F16K (31/535); F16K (31/60); F16K (1/2268); F16K (5/0694); B64D (37/30); B64D (37/005); B64F (1/28); E04H (6/44)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2906143	12/1958	Walton	74/625	F16H 49/001
4616529	12/1985	Heinemann	74/640	F16H 49/001
4951518	12/1989	Hendershot	N/A	N/A
9506571	12/2015	Mccormack	N/A	N/A
10871213	12/2019	Ishikawa	N/A	F16H 55/08
2015/0292400	12/2014	Hiyoshi	123/48R	F02B 75/045
2015/0336680	12/2014	Schumacher et al.	N/A	N/A
2017/0341769	12/2016	Haberbusch et al.	N/A	N/A
2021/0078702	12/2020	Ciet et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102014107316	12/2014	DE	N/A
102019212221	12/2020	DE	N/A
2021148335	12/2020	WO	N/A

OTHER PUBLICATIONS

European Search Report for corresponding European Patent Application No. 22204327 dated Apr. 18, 2023; priority document. cited by applicant

Wikipedia, Strain wave gearing, download on Oct. 25, 2022 from https://en.wikipedia.org/wiki/Strain_wave_gearing. cited by applicant

Primary Examiner: Bastianelli; John

Attorney, Agent or Firm: Greer, Burns & Crain, Ltd.

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of European patent application No. 22204327.5 filed on Oct. 28, 2022, the entire disclosures of which are incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The invention relates to a valve actuation mechanism for actuating a valve for cryogenic applications and/or H2 applications. Further, the invention relates to a valve arrangement having such a valve actuating mechanism. Further, the invention relates to a cryogenic pipe installation for conducting liquid hydrogen, a vehicle, and an airport as possible uses of such valve arrangement.

BACKGROUND OF THE INVENTION

(3) For technical background, reference is made to the following citations:

(4) [1] DE 10 2014 107 316 A1

(5) [2] US 2021/0 078 702A1

(6) [3] WO 2021/148 335A1

(7) [4] Wikipedia, Strain wave gearing, download on 25.10.2022 from https://en.wikipedia.org/wiki/Strain_wave_gearing

(8) [1] to [3] relate to hydrogen installations in an aircraft.

SUMMARY OF THE INVENTION

(9) Preferred embodiments of the invention relate to parts of installations of hydrogen systems on an aircraft, vehicles or on the ground. When an aircraft has at least one hydrogen consumer, such as a fuel cell or an engine powered at least partially with hydrogen, a hydrogen distribution system is needed on the aircraft. The hydrogen distribution system especially includes a hydrogen duct system with pipes, couplings, pipe branches, equipment and so on for distributing the hydrogen, which may be liquid hydrogen (LH2, especially cryogenic) or gaseous hydrogen (GH2). Further, for fueling aircraft powered with hydrogen, hydrogen systems are also needed on airports and eventually on tank vehicles. Especially, valves are needed to control, enable or stop flow of hydrogen.

(10) An object of the invention is to improve a valve actuation mechanism for use with high anti-leakage requirements, especially in H2 applications, more particularly LH2 applications.

(11) The invention provides a valve actuation mechanism for actuating a rotatable closing element of a valve for cryogenic applications and/or H2 applications, comprising:

(12) a housing defining an inner area which is sealed by a sealing membrane from an outer area,

(13) an input shaft for receiving a rotational movement wherein the input shaft is arranged in the outer area,

(14) an output shaft to be connected to the closing element, wherein the output shaft is arranged in the inner area, and

(15) a strain wave gearing connecting the input shaft and the output shaft, wherein the strain wave gearing comprises a wave generator connected to one of the input and output shafts, a rotatable outer gear wheel connected to the other of the input and output shafts and a stationary flexible ring gear with external teeth which is deformed by the wave generator to engage with internal gear teeth of the outer gear wheel, wherein the flexible ring gear is provided as a portion of the sealing membrane.

(16) Preferably, the input shaft is selected from the group consisting of a shaft of a manual lever, a driving shaft of a motorized actuator, a driving shaft of an electric actuator, a driving shaft of a pneumatic actuator, a shaft of a crank, a shaft of a crank lever, a shaft of a handwheel.

(17) Preferably, the housing is a canisterized housing with an internal vacuum.
(18) Preferably, the housing defines the inner area as a first sealed chamber and further defines a second sealed chamber sealed by the sealing membrane from the first sealed chamber, wherein at least a portion of the input shaft connected to one of the wave generator and the outer gear wheel is arranged within the second sealed chamber.
(19) Preferably, the housing has a connection means to be connected to a valve housing of a valve to be actuated.
(20) Preferably, the housing has a bolted removable cap with static seal.
(21) Preferably, the membrane seal comprises a pot shaped center area with an annular wall wherein at least an annular portion of the annular wall comprises the external teeth of the flexible ring gear.
(22) Preferably, the membrane seal comprises an annular intermediate area extending essentially radially with a wave shaped expansion portion, especially a bellow shaped portion.
(23) Preferably, the membrane seal comprises an outer rim portion or flange portion fixedly connected to the housing.
(24) According to another aspect, the invention provides a valve arrangement for LH2 applications, comprising a valve with a closing element which is movable between a closed position and an open position by a rotation movement and a valve actuation mechanism according to any of the aforementioned embodiments.
(25) Preferably, the valve is a ball valve or butterfly valve.
(26) Preferably, the valve arrangement comprises a vacuum capsule boundary, wherein the housing is welded to the capsule.
(27) According to another aspect, the invention provides a cryogenic pipe installation for conducting liquid hydrogen, comprising a valve arrangement according to any of the aforementioned embodiments.
(28) According to another aspect, the invention provides a vehicle, especially an aircraft with at least one hydrogen consumer or a LH2 tank vehicle, comprising a valve actuating mechanism, a valve arrangement, and/or a cryogenic pipe installation according to any of the aforementioned embodiments.
(29) According to another aspect, the invention provides an airport comprising a cryogenic pipe installation with such valve actuation mechanism according to any of the aforementioned embodiments.
(30) Preferred embodiments can be used in the technical field of small molecule gas transportation to be controlled via valves subjected to very high leakage tightness requirements.
(31) Preferred embodiments relate to gas tight membrane seals for rotation transmission. Preferred embodiments relate to a gas tight rotational seal. Preferred embodiments relate to hydrogen control and distribution equipment.
(32) The invention especially lies on the technical field of hydrogen powered aircraft and installations related therewith. One of the challenges connected therewith is the integration of hydrogen control and distribution equipment. Shut-off and distribution ball valves are the preferred choice over piston valves or poppet valves. A challenge experienced with use of ball valves is that they require a rotational seal. It is quite difficult to provide a rotational seal such that it is gas tight for liquid hydrogen use. Thus, any dynamic movements would increase leakage. Leaking of larger amounts of H2 should be avoided for different reasons. This phenomenon is not present in a membrane sealed piston poppet valve due to the translational operation of the closing elements thereof.
(33) Embodiments of the invention make use of a concept similar to a strain wave gearing (also called “harmonic drive”) as known from reference [4]. In embodiments of the invention, a membrane transmits a torque moment to separate components involved in the mechanism.
(34) In usual harmonic drives as known from reference [4], the outer-toothed wheel is usually

fixed. This allows an inner ellipsoid gear, plus the toothed membrane to rotate. Similar to a planetary gear, the setup of the two moving parts around one steady one allows for simple maintenance.
(35) According to embodiments of the invention, the flexible gear is formed by a part of a sealing membrane. The sealing membrane is stationary so that it can be tightly fixed within a housing, and it can securely seal an inner area of the housing from an outer area. Other as in usual harmonic drives the flexible gear, as part of the sealing membrane is fixed, while the outer wheel is rotatable relative to the housing. The wave generator has a rotatable element such as an ellipsoid gear or any other eccentric gear that is suitable to deform the flexible gear such that some of the outer teeth thereof engage some of the inner teeth of the outer wheel. Hence, rotational movement can be transmitted via gas tight stationary sealing.
(36) Either the wave generator or the outer wheel could be used as input while the other thereof is used as output for actuating a valve. In preferred embodiments, the rotatable element of the wave generator is used as input for the rotational movement, and the outer wheel is used as output that can be connected to a rotating closing element of a valve such as a ball valve.
(37) Ball valve shut-off valves have some advantages over piston valves for LH2 applications, particularly for the non-motorized ones. The challenge is that the actuation of the ball valve requires a shaft rotation which—as far as known—today cannot be designed leakage tight for very small molecular gases in a similar way as this is possible with a longitudinal actuation for a piston valve to be sealed by a membrane.
(38) Embodiments of the invention make use of the principle of harmonic drive gear boxes such as known from reference [4] for transmitting rotational movements for actuating valves. This is especially useful for ball valves but of course also for all other valves that can be actuated—directly or indirectly—by a rotational movement.
(39) One drawback could be that harmonic drive gearboxes usually have a transmission ratio more than 1:25 so that one needs more than six turns to close a ball valve with a 90° movement. Therefore, manual operation elements that easily can be turned several times are preferred such as cranks, hand wheels or crank levers. Of course, automatic rotation drive actuators such as stepper motors, electric actuators, hydraulic motors or pneumatic motors could also be connected to the input shaft.
(40) Since there is no relative movement in the seal but the membrane uses its elastic deformation behavior, the achievable leakage rate is extremely low.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the invention are explained below referring to the accompanying drawings in which:
(2) FIG. **1** is a schematic sectional view of a valve actuation mechanism according to an embodiment of the invention;
(3) FIG. **2** is a sectional view along line II-II of FIG. **1**;
(4) FIG. **3** is a schematic sectional view of a valve arrangement comprising a valve actuation mechanism according to an embodiment of the invention;
(5) FIG. **4** is a schematic perspective view of an aircraft comprising a valve arrangement of FIG. **3**;
(6) FIG. **5** is a schematic view of a tank vehicle comprising a valve arrangement of FIG. **3**; and
(7) FIG. **6** is a schematic view of an airport comprising a valve arrangement of FIG. **3**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(8) FIG. **1** shows a valve actuation mechanism **10** for actuating a valve **12** for cryogenic applications and/or H2 applications. FIG. **2** shows a section through parts of the valve actuation

mechanism **10** along line II-II of FIG. **1**. FIG. **3** shows a valve arrangement **14** including a valve **12** and the valve actuation mechanism **10** for actuating the valve **12**.
(9) The valve actuation mechanism **10** is configured to actuate a rotatable closing element **16** of the valve **12** in cryogenic applications and/or H2 applications. Especially, the valve arrangement **14** comprising the valve **12** and the valve actuation mechanism **10** is configured to control flow of liquid hydrogen in a cryogenic pipe installation **18** such as in a hydrogen distribution system **20**.
(10) Referring to FIGS. **1** and **2**, the valve actuation mechanism **10** comprises a housing **22**, a sealing membrane **24**, an input shaft **26**, an output shaft **28**, and a strain wave gearing **30**.
(11) The housing **22** defines an inner area **32** sealed by the sealing membrane **24** from an outer area **34**.
(12) The input shaft **26** is configured to receive a rotational movement especially from a manually operated element such as a handle, a crank or a hand wheel (not shown). The input shaft **26** is arranged in the outer area **34**.
(13) The output shaft **28** is configured to be connected to the closing element **16** and is arranged in the inner area **32**.
(14) The strain wave gearing **30** connects the input shaft **26** and the output shaft **28** and transmits a rotational movement **36** from the input shaft **26** to the output shaft **28**.
(15) The strain wave gearing **30** comprises a wave generator **38** connected to one of the input and output shafts **26**, **28**, a rotatable outer gear wheel **40** connected to the other of the input and output shafts **26**, **28** and a stationary flexible ring gear **42** with external teeth **44**. The flexible ring gear **42** is deformed by the wave generator **38** to engage with internal gear teeth **46** of the outer gear wheel **40**. According to embodiments of the invention, the flexible ring gear **42** is provided by a portion of the sealing membrane **24**. In other words, a ring portion **48** of the membrane **24** is provided with the external teeth **44** and acts as the flexible ring gear **42** of the strain wave gearing **30**.
(16) The input shaft **26** can be, e.g., a shaft **50** of a manual lever, a driving shaft of a motorized actuator **52**, a driving shaft of an electric actuator, a driving shaft of a pneumatic actuator, a shaft of a crank, a shaft of a crank lever, or a shaft of a handwheel.
(17) In some embodiments, the housing **22** is configured as a sealed canister with a thermal insulation, especially with vacuum **54**. In some embodiments, the housing **22** has a first sealed chamber **56** and a second sealed chamber **58**. The chambers **56**, **58** are sealed from each other by the membrane **24**. The first sealed chamber **56** constitutes at least a part of the inner area **32** and encloses the output shaft **28**. The second sealed chamber **58** forms a part of the outer area **34**, wherein at least a portion of the input shaft **26** connected to the one of the wave generator **38** and the outer gear wheel **40** is arranged within the second sealed chamber **58**.
(18) Referring to FIG. **3**, the housing **22** can be part of a valve housing **60** of the valve **12**. According to some embodiments, the housing **22** is or has a removable screwed section of the valve housing **60**, wherein the screw connection is sealed by a static seal. When connected, the first sealing chamber **56** receives a driven end **62** of the closing element **16**. In some embodiments, the valve **12** is a ball valve or butterfly valve, and the closing element **16** comprises a rotating ball element or butterfly. The driven end **62** is connected for common rotation to the output shaft **28**.
(19) Referring to FIGS. **1** to **3**, the membrane **24** comprises an outer rim portion **64** or flange portion fixedly connected to the housing **22**. Especially, the rim portion **64** is clamped in a sealing manner between an inner flange **66** in the interior of the housing **22** and a clamping ring **68**.
(20) Further, the membrane **24** comprises a dome shaped or pot shaped center area **70** with an annular wall wherein at least an annular portion of the annular wall constitutes the ring portion **48** with the external teeth **44** as the flexible ring gear **42**.
(21) In some embodiments, the membrane **24** comprises an annular intermediate area **72** extending essentially radially with a wave shaped expansion portion. The intermediate area **72** is formed as a bellow **74** or has at least a bellow section. The intermediate area connects the pot shaped center area **70** to the rim portion **64** and provides good flexibility for an eccentric movement of the pot

shaped portion **70** during a deformation induced by the wave generator **38**. In some embodiments, the membrane **24** is made of metal.

(22) In embodiments not shown, the wave generator **38** is connected to the output shaft **28**, and the outer gear wheel **40** is connected to the input shaft **26**. In the preferred embodiments as shown, the wave generator **38** is connected to the input shaft **26** and the outer gear wheel **40** is connected to the output shaft, and hence, to the closing element **16** of the valve **12**.

(23) The strain wave gearing **30** (also known as harmonic gearing) is a type of mechanical gear system that uses the flexible ring gear **42** (also called flexible spline) with external teeth **44**, which is deformed by a rotating elliptical plug **76** to engage with the internal gear teeth **46** of the outer gear wheel **40** (also called outer spline).

(24) The wave generator **38** comprises an elliptical disk called a wave generator plug **76** and an outer ball bearing **78**. The elliptical plug **76** is inserted into the ball bearing **78**, forcing the bearing **78** to conform to the elliptical shape but still allowing rotation of the plug **76** within the outer bearing **78**.

(25) The membrane **24** functioning as a seal and as a flexible spline is shaped like a shallow cup. Due to the form of the membrane **24**, the membrane wall at the ring portion **48** is very flexible while the membrane **24** is rigid enough to be tightly secured to the housing in order to withstand torque induced therein. Teeth **44** are positioned radially around the outside of the flexible ring portion **48**. The ring portion **48** fits tightly over the wave generator **38**, so that when the wave generator plug **76** is rotated, the ring portion **48** deforms to the shape of a rotating ellipse and does not slip over the outer elliptical ring of the ball bearing **78**. The ball bearing enables a smooth rotation of the wave generator **38** within the ring portion **48**.

(26) FIG. **3** shows the valve arrangement **14** with a ball valve as valve **12**. The valve **12** is enclosed in a vacuum capsule **80** boundary. Dynamic seals **82** comprise seal elements of the valve **12** rotating relative to each other. A medium such as LH2 leaking through the dynamic seals **82** can only leak into the inner area **32**, e.g., the first sealed chamber **56**, but is prevented from leaking into the second sealed chamber **58** or other parts of the outer area **34** by means of the sealing membrane **24**.

(27) The valve **12** is configured as a shut-off ventil in a cryogenic pipe installation **18**. The valve **12** comprises pipe sections **84** connected to hoses **86** or other ducts of the pipe installation **18**. The valve **12** is spaced apart from walls of the capsule **80**, e.g., by means of valve supports **88**.

(28) Connections between the pipe sections **84** and the valve housing **60**, and between the canister, i.e., the housing **22** of the valve actuation mechanism **10** and the capsule are made with gas tight weld connections **90**.

(29) The second sealed chamber **58** is closed by a bolted removable cap **92** with static seal. A vacuum port **93** is provided, either as shown in the cap **92** or in the housing wall enclosing the second sealed chamber **58** in order to establish a vacuum in the second sealed chamber **58**.

(30) With the valve actuation mechanism **10**, the vacuum integrity of the capsule **80** can be maintained, and rotational movement can be transmitted without the need of any dynamic seal. Hence, a very high safety against leakage of H2 is achieved.

(31) As shown in FIGS. **4**, **5**, and **6**, the cryogenic pipe installation **18** can be used in an aircraft **94** for controlling flow and distribution of LH2 from and to LH2 tanks **96**, on board of a tank vehicle **98**, for example for fueling aircraft, and on an airport **100**, for example for distributing LH2 as aircraft fuel to aircraft parking positions **102**.

(32) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or

both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.
REFERENCE SIGN LIST
(33) **10** valve actuation mechanism **12** valve **14** valve arrangement **16** closing element **18** cryogenic pipe installation **20** hydrogen distribution system **22** housing **24** membrane **26** input shaft **28** output shaft **30** strain wave gearing **32** inner area **34** outer area **36** rotational movement **38** wave generator **40** outer gear wheel **42** flexible ring gear **44** external teeth **46** internal gear teeth **48** ring portion **50** shaft of a manual lever **52** actuator **54** vacuum **56** first sealed chamber **58** second sealed chamber **60** valve housing **62** driven end **64** rim portion **66** inner flange **68** clamping ring **70** pot shaped center area **72** intermediate area **74** bellow **76** elliptical plug **78** bearing **80** vacuum capsule **82** dynamic seal **84** pipe section **86** hose **88** valve support **90** weld connection **92** cap **93** vacuum port **94** aircraft **96** LH2 tank **98** tank vehicle **100** airport **102** parking position

Claims

1. A valve actuation mechanism for actuating a rotatable closing element of a valve for at least one of cryogenic applications or hydrogen applications, comprising: a housing defining an inner area which is sealed by a sealing membrane from an outer area, an input shaft configured to receive a rotational movement wherein the input shaft is arranged in the outer area, an output shaft configured to be connected to the closing element, wherein the output shaft is arranged in the inner area, and a strain wave gearing connecting the input shaft and the output shaft, wherein the strain wave gearing comprises a wave generator connected to one of the input and output shafts, a rotatable outer gear wheel connected to the other one of the input and output shafts and a stationary flexible ring gear with external teeth which is deformed by the wave generator to engage with internal gear teeth of the outer gear wheel, wherein the flexible ring gear is provided as a portion of the sealing membrane, wherein the sealing membrane comprises a pot shaped center area with an annular wall wherein at least a ring portion of the annular wall comprises the external teeth of the flexible ring gear.
2. A valve actuation mechanism according to claim 1, wherein the input shaft is selected from the group consisting of a shaft of a manual lever, a driving shaft of a motorized actuator, a driving shaft of an electric actuator, a driving shaft of a pneumatic actuator, a shaft of a crank, a shaft of a crank lever, and a shaft of a handwheel.
3. The valve actuation mechanism according to claim 1, wherein the housing comprises a canisterized housing with an internal vacuum.
4. The valve actuation mechanism according to claim 1, wherein the housing, defines the inner area as a first sealed chamber and further defines a second sealed chamber sealed by the sealing membrane from the first sealed chamber, wherein at least a portion of the input shaft and the one of the wave generator and the outer gear wheel connected thereto are arranged within the second sealed chamber.
5. The valve actuation mechanism according to claim 1, wherein the housing has a connection arrangement configured to be connected to a valve housing of a valve to be actuated.
6. The valve actuation mechanism according to claim 1, wherein the housing has a bolted removable cap with a static seal.
7. The valve actuation mechanism according to claim 1, wherein the sealing membrane comprises a deformable, bellow shaped, annular intermediate area extending essentially radially.
8. The valve actuation mechanism according to claim 1, wherein the sealing membrane comprises an outer rim portion or flange portion fixedly connected to the housing.
9. A valve arrangement for liquid hydrogen applications, comprising: a valve with a closing

element which is movable between a closed position and an open position by a rotational movement, and a valve actuation mechanism according to claim 1.
10. The valve arrangement according to claim 9, wherein the valve is a ball valve or butterfly valve.
11. The valve arrangement according to claim 9, comprising a vacuum capsule boundary, wherein the housing is welded to the vacuum capsule.
12. A cryogenic pipe installation for conduction liquid hydrogen, comprising a valve arrangement according to claim 9.
13. A vehicle with at least one hydrogen consumer or liquid hydrogen tank, comprising a valve actuation mechanism according to claim 1.
14. A vehicle with at least one hydrogen consumer or liquid hydrogen tank, comprising a valve arrangement according to claim 9.
15. A vehicle with at least one hydrogen consumer or liquid hydrogen tank, comprising a cryogenic pipe installation according to claim 12.
16. An aircraft with at least one hydrogen consumer or liquid hydrogen tank, comprising a valve actuation mechanism according to claim 1.
17. An aircraft with at least one hydrogen consumer or liquid hydrogen tank, comprising a valve arrangement according to claim 9.
18. An aircraft with at least one hydrogen consumer or liquid hydrogen tank, comprising a cryogenic pipe installation according to claim 12.
19. An airport comprising a cryogenic pipe installation according to claim 12.
20. A valve actuation mechanism for actuating a rotatable closing element of a valve for at least one of cryogenic applications or hydrogen applications, comprising: a housing defining an inner area which is sealed by a sealing membrane from an outer area, wherein the housing comprises a canisterized housing with an internal vacuum, an input shaft configured to receive a rotational movement wherein the input shaft is arranged in the outer area, an output shaft configured to be connected to the closing element, wherein the output shaft is arranged in the inner area, and a strain wave gearing connecting the input shaft and the output shaft, wherein the strain wave gearing comprises a wave generator connected to one of the input and output shafts, a rotatable outer gear wheel connected to the other one of the input and output shafts and a stationary flexible ring gear with external teeth which is deformed by the wave generator to engage with internal gear teeth of the outer gear wheel, wherein the flexible ring gear is provided as a portion of the sealing membrane.